

Towards a Virtual Reality-Based Educational System for Teaching Single-Layer Perceptron Concepts

Navid Khezrian*

Forouzan Farzinnejad†

Jens Grubert‡

Coburg University of Applied Sciences and Arts

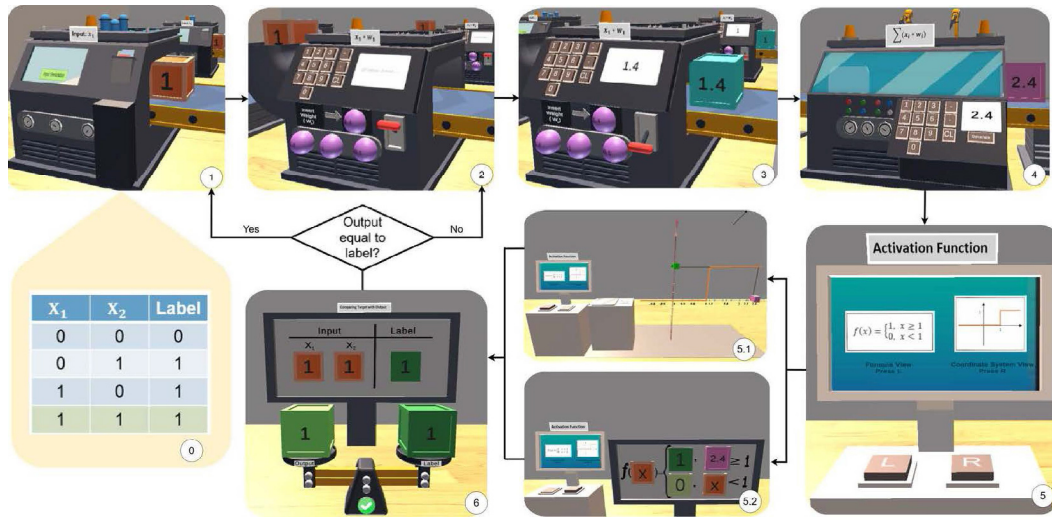


Figure 1: Overview of the VR-based educational system for teaching single-layer perceptron Concepts via a logical OR gate use case: 0) Defining OR gate states 1) Generating random inputs 2) Applying inputs to a multiplication unit 3) Assigning weights and calculating products 4) Summing outputs in a summation unit 5) Activation function display with graph (5.1) and formula (5.2) visualization options 6) Evaluation of the perceptron output with the ground truth label.

ABSTRACT

We present a Virtual Reality (VR) educational system for teaching single-layer perceptron concepts, offering an immersive and interactive approach to enhance traditional learning methods. The VR platform gamifies neural network training by simulating a logical OR gate in a factory-inspired environment, aiming at bridging abstract theories with practical applications. In a user study with eight participants, the system received positive but also critical feedback for its interactivity and usability, achieving a System Usability Scale score of 64.69. As an initial study on teaching machine learning in VR, this work highlights VR's potential to enhance comprehension and engagement through experiences.

Index Terms: Virtual Reality, Machine Learning Education, Single-Layer Perceptron, Immersive Learning

1 INTRODUCTION

Virtual Reality (VR) technology has emerged as a transformative tool in education, particularly within computer science and engineering disciplines, due to its ability to enhance user engagement, motivation, and comprehension through immersive, interactive environments [1, 5, 6]. Research has shown that VR can provide ef-

fective and efficient learning experiences by incorporating game-based learning and gamification techniques, especially in computer science education [1]. Moreover, teaching through playing is a branch of gamification that enhances the learning process by fostering greater engagement, enjoyment, and interactivity [4]. Through virtual experiences, VR enables students to engage in practical experiments that may otherwise be too expensive, dangerous, or time-consuming to conduct in a traditional classroom setting, thus making it highly beneficial in science, technology, engineering, and mathematics (STEM) education [6]. Furthermore, VR fosters a strong sense of presence, contributing to improved learning outcomes and supporting the development of mental models for computer science concepts [3, 6].

Based on these insights, we propose a VR-based educational system designed to enhance the understanding of single-layer perceptron concepts through an interactive and immersive experience. By incorporating gamification elements and simulating the implementation of a logical OR gate in a factory-inspired virtual environment, the system bridges abstract computational theories with hands-on, practical learning. To assess the system's efficacy in improving learners' comprehension and engagement, we conducted an initial user study, focusing on its impact on usability, and user experience.

2 SYSTEM DESIGN AND IMPLEMENTATION

The VR-based system teaches single-layer perceptron concepts, guiding learners step-by-step through the feed-forward process of a logical OR gate. Designed for machine learning beginners, the system uses gamification and hands-on tasks to make abstract concepts tangible. Learners actively engage with stations that visually represent computational steps as part of a production line, using inter-

*e-mail: navid.khezrian@hs-coburg.de

†e-mail: forouzan.farzinnejad@hs-coburg.de

‡e-mail: jens.grubert@hs-coburg.de

active elements like blocks and machinery to connect mathematical processes with physical actions, fostering deeper understanding and engagement. For navigation, we employed teleportation facilitated by raycasting, while interaction with objects, such as grabbing, was achieved using controllers.

The feed-forward process is organized into clear steps, as illustrated in Fig.1, to understand how the single-layer perceptron works. First, the input states for the OR gate are defined (Step 0), and random inputs are generated (Step 1). These inputs go to the multiplication unit (Step 2), where weights are assigned, and the products are manually calculated and entered into the system (Step 3). The calculated products are then sent to the summation unit (Step 4) to combine them into a single value. This value is passed to the activation function module (Step 5), where users choose how to represent the function (as a formula or graph). The activation function is applied, producing the perceptron's binary output (1 or 0) (Steps 5.1 or 5.2). The output is compared to the expected target (Step 6). If the output is correct, the process repeats for all input states (Steps 1–6). If it's incorrect, weights are adjusted at Step 2, and the steps are repeated until the perceptron produces correct outputs for all inputs. This iterative process fine-tunes the weights, allowing the perceptron to accurately model the OR gate's behavior.

The VR system was built with the Unity game engine (version 2022.3.5f1), using the XR Interaction Toolkit, and deployed on Meta Quest 3 head-mounted displays with controller-based interactions. This VR system incorporates an interactive, factory-inspired design that aims to facilitate an engaging and immersive learning environment. By employing gamification techniques, the platform seeks to present neural network concepts in a structured and accessible manner.

3 EVALUATION

The VR system was initially evaluated with eight participants, including seven males and one female, aged 25.63 on average (SD = 2.97). The target population for the system was university students majoring in computer science. All had XR experience, and six with AI backgrounds provided usability and user experience insights.

The evaluation commenced with an overview of the VR system and task objectives, followed by an introduction to prerequisite concepts and instructions for interacting with VR objects. Participants then completed VR training and provided qualitative feedback and usability ratings via the System Usability Scale (SUS) [2]. IRB approval was not required for this study at our university.

Each session, including the briefing, VR task performance, and feedback collection, lasted approximately 40 minutes per participant. For the sake of this initial evaluation, all participants focused on using the formula view (5.2). The qualitative feedback highlighted several positive aspects of the VR system, specifically regarding its intuitive design, interactive elements, and potential for facilitating enjoyable learning experiences.

The qualitative feedback indicated that: 1) Five participants valued the environment's interactivity, noting it made AI learning intuitive and enjoyable. One highlighted, "It's great to see perceptron progress in action, It's cool that it is so interactive." 2) Participants had mixed feelings about moving between stations. Three found the teleportation easy to use, while four found it challenging. One participant suggested implementing joystick controls or walking options in the future, citing confusion and disorientation. He noted, "You accidentally teleport and don't know where you are." 3) Three participants suggested integrating theoretical educational materials into the VR environment to enhance learning. One participant remarked, "Having the concepts within the VR environment would help us learn better by studying them while working with the objects." 4) Four Participants emphasized the need for clear, accessible interaction guidelines, as initial instructions were hard to recall during use. One noted, "I forgot how to interact with objects

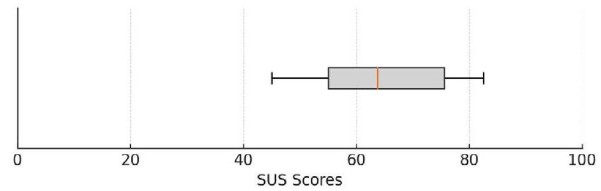


Figure 2: Boxplot of the system usability scale scores.

while doing it."

The VR system demonstrated moderate usability, with an average SUS score of 64.69 (SD = 16.62; see Figure 2), indicating acceptable but improvable usability. The variability suggests differing user experiences, which is in line with the participants' comments. This also highlights areas for optimization to enhance consistency and overall satisfaction in future iterations of the system.

4 CONCLUSION AND FUTURE WORK

We presented a Virtual Reality-based educational platform specifically designed to enhance comprehension of single-layer perceptron concepts through an immersive and gamified learning experience. Evaluation results demonstrated the system's ability to make AI concepts intuitive and engaging, with participants praising its interactivity and design. However, a moderate SUS score of 64.69 and feedback also indicated the need for further improvements, including navigation, interaction guidelines, and integrating theoretical materials.

The current study primarily focused on evaluating the overall usability of the VR system, providing a foundation for future enhancements. Building on these findings, future work will focus on refining the system to improve usability, making it more inclusive for a broader range of users. Additionally, a detailed analysis will be conducted to explore correlations between participants' backgrounds and their SUS scores, enhancing the robustness and depth of the findings. The system's efficacy in promoting student learning will also be evaluated through comparative analyses with traditional machine learning training methods, such as Jupyter Notebooks or instructional booklets, to further validate its educational effectiveness and impact.

REFERENCES

- [1] F. J. Agbo, I. T. Sanusi, S. S. Oyelere, and J. Suhonen. Application of virtual reality in computer science education: A systemic review based on bibliometric and content analysis methods. *Education Sciences*, 11(3):142, 2021. 1
- [2] J. Brooke et al. Sus-a quick and dirty usability scale. *Usability evaluation in industry*, 189(194):4–7, 1996. 2
- [3] O. E. Cinar, K. Rafferty, D. Cutting, and H. Wang. A comparative study: Augmented and virtual reality applications for improving comprehension of abstract programming concepts. In *VISIGRAPP (1): GRAPP, HUCAPP, IVAPP*, pp. 538–548, 2024. 1
- [4] F. Farzinnejad, J. Rasti, N. Khezrian, and J. Grubert. The effect of an exergame on the shadow play skill based on muscle memory for young female participants: The case of forehand drive in table tennis. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 1152–1160. IEEE, 2023. 1
- [5] D. López-Fernández, J. Mayor, J. Pérez, and A. Gordillo. Learning and motivational impact of using a virtual reality serious video game to learn scrum. *IEEE Transactions on Games*, 15(3):430–439, 2022. 1
- [6] J. Pirker, A. Dengel, M. Holly, and S. Safikhani. Virtual reality in computer science education: A systematic review. In *Proceedings of the 26th ACM symposium on virtual reality software and technology*, pp. 1–8, 2020. 1